

ANN7: XOR using Backpropagation – Detailed Code & Viva Explanation

Cell 1: Import Library

Code:

```
import numpy as np
```

Explanation (what to say in viva):

Here I am importing numpy library for numerical operations.

It is required for matrix calculations in neural networks.

Cell 2: Define Dataset

Code:

```
x = np.array([[0,0],[0,1],[1,0],[1,1]])  
y = np.array([[0],[1],[1],[0]])
```

Explanation (what to say in viva):

X contains input combinations for XOR.

y contains expected outputs.

XOR is a non-linear problem.

It cannot be solved using single layer perceptron.

Cell 3: Initialize Weights

Code:

```
w1 = np.random.rand(2,2)  
w2 = np.random.rand(2,1)
```

Explanation (what to say in viva):

W1 connects input layer to hidden layer.

W2 connects hidden layer to output layer.

Random initialization helps in proper learning.

Cell 4: Activation Function

Code:

```
def sigmoid(x):  
    return 1/(1+np.exp(-x))
```

Explanation (what to say in viva):

Sigmoid converts input to range 0 to 1.

It introduces non-linearity.

Essential for solving XOR problem.

Cell 5: Forward Pass

Code:

```
z1 = np.dot(X, W1)
a1 = sigmoid(z1)
z2 = np.dot(a1, W2)
a2 = sigmoid(z2)
```

Explanation (what to say in viva):

z1 is input to hidden layer.

a1 is output of hidden layer.

z2 is input to output layer.

a2 is final output.

This process is forward propagation.

Cell 6: Error Calculation

Code:

```
error = y - a2
```

Explanation (what to say in viva):

This calculates difference between actual and predicted output.

It helps in improving model performance.

Cell 7: Backpropagation

Code:

```
dw2 = np.dot(a1.T, error)
dw1 = np.dot(X.T, error)
```

Explanation (what to say in viva):

Gradients are computed to adjust weights.

Error is propagated backward.

This is key step in learning.

Cell 8: Update Weights

Code:

```
w1 += dw1
w2 += dw2
```

Explanation (what to say in viva):

Weights are updated using gradients.

This reduces error over time.

Cell 9: Output

Code:

```
print(a2)
```

Explanation (what to say in viva):

This prints final output.

Output should match XOR truth table after training.

